

A Two-Stage Diffusion Pipeline for Consistent Front-Back Clothing Generation

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ABSTRACT

This study introduces a novel pipeline to generate consistent front and back apparel image datasets, addressing the limitations of current virtual try-on (VTON) data. Our approach first uses the Llama3-Instruct model, integrating the chain-of-thought (CoT) and instruction tuning, to convert simple user text into detailed and attribute-consistent textual descriptions for front and back garment views. A subsequent two-stage diffusion process generates the images: a flux model creates the front view, and then an SDXL model, combined with an IP-Adapter that references the front image, produces the corresponding back view. This methodology demonstrably enhanced visual consistency and details of the resulting high-resolution image pairs. Through qualitative and quantitative evaluations, this study validated the practical applicability of the pipeline and highlighted its potential to significantly improve data diversity and quality for applications such as VTON, fashion design, and 3D modeling.

Key words : Chain of Thought, Instruct Tuning, Llama, Large Language Model (LLM), VTON

1. Introduction

Various approaches have emerged in the field of artificial intelligence to develop models capable of both generating high-quality data and solving complex problems. The field of image generation has advanced through innovations in deep learning-based generative models. Prominent examples include Varia-

tional Autoencoders (VAEs) and Generative Adversarial Networks (GANs). A VAE encodes input data into a latent vector within a continuous probability distribution, then generates new data through sampling. In this process, the VAE effectively learns the latent structure of the given data distribution to produce similar examples. While VAEs can efficiently capture the latent representation of complex data, their generated images tend to be somewhat blurry. GANs were subsequently introduced to address this limitation of VAEs [1,2]. GANs incorporate an adversarial structure where a generator and a discriminator compete during training. Through this, GANs have succeeded in producing images so clear and realistic that they are difficult to distinguish from real ones [3,4]. This innovation has significantly raised the standard for image synthesis quality and expanded the range of deep learning applications [5]. More recently, various architectural modifications and training techniques have been developed to enable high-resolution image

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generation and finer control in GANs. Furthermore, ideas derived from VAEs and GANs are also being integrated into new paradigms like diffusion models [6]. These evolved generative models have found applications in areas such as Virtual Try-On (VTON) [7], making it possible to realistically composite images in which a user's photo is dressed in the desired clothing. For example, POVNet, a state-of-the-art VTON model, employs precise warping and partial image generation techniques to preserve the texture of garments when naturally compositing them onto a person's image. Through adversarial training, it represents even fine details with photorealistic precision, offering improved realism compared to previous methods [8]. As a result, subtle folds and shading are accurately reproduced in the synthesized images, enabling a realistic on-line fitting experience in e-commerce settings and leading to reported increases in user engagement.

However, existing research has primarily focused on analyzing the effects of prompt structures and hyperparameters in text-to-image generation models [9], or on utilizing automatic recommendation systems to aid users in their exploration processes [10]. Previous studies have primarily focused on text-to-image transformation and optimization. However, the significance of this research lies in its progression towards constructing a consistent dataset that incorporates logical reasoning. Earlier approaches typically relied on straightforward transformations or matching of user input. In contrast, this study employs Chain of Thought (CoT) [11] to generate refined prompts through a logical intermediate inference process. Furthermore, by combining Instruction Tuning [12] with CoT, it becomes possible to produce prompts that contain far more comprehensive information than conventional methods. Specifically, we propose a strategy employing CoT to logically link attributes within the textual descriptions for front and back views. Instruction Tuning is used to automatically expand initial user prompts, thereby creating more precise and consistent textual inputs for dataset generation. This method holds the potential for designing AI models that maintain data consistency across various industries. In other words, this research goes beyond merely developing superior prompts; it offers a new paradigm wherein AI applies logical reasoning processes to construct practically usable datasets, setting it apart in terms of originality and effectiveness.

In summary, there are two leading trends in recent AI research. One is the advancement in image and content generation technologies (e.g., VAEs, GANs, and their applications like VTON). The other is the improvement in reasoning and adapt-

ability based on CoT and Instruction Tuning. On one hand, high-quality image generation is opening new opportunities in fields such as virtual reality and content creation. On the other hand, models can now faithfully follow user instructions and logically solve complex tasks, dramatically elevating the performance of intelligent systems that interact with humans. Ultimately, the convergence of these technologies is making it increasingly feasible to develop next-generation AI systems equipped with both high expressive power and sophisticated reasoning abilities. Within this context, the present study aims to incorporate the latest techniques into the fashion image generation problem. The specific goal of this study is to address the challenge of constructing a high-quality dataset of apparel images that exhibit strong consistency between their front and back views. To this end, we combine CoT and Instruction Tuning to generate detailed prompts for both front and back images from a simple user text input. Based on this, we propose a novel apparel image generation pipeline.

This paper is organized as follows. Chapter 2 discusses related research and existing methodologies in detail, and Chapter 3 presents the theoretical background and implementation details of the proposed approach. In Chapter 4, we validate the effectiveness of our method through experiments, presenting both qualitative and quantitative evaluations, and Chapters 5 and 6 provide conclusions and discuss directions for future research.

2. Related Work

In this section, we review prior studies closely related to the proposed methodology. We summarize diffusion models in image generation, high-resolution diffusion models (SDXL), the Flux model, and the IP Adapter employed in our proposed approach. We also introduce Large Language Models (LLMs) in the field of natural language processing, along with Instruction Tuning and the CoT technique. By covering these related works, we clarify the theoretical foundations upon which this paper's approach is built.

2.1. Diffusion model

Diffusion refers to a process in which specific patterns or structures in data gradually disappear, ultimately turning into random noise. The Diffusion Model [13,14] leverages this diffusion concept and consists of two stages: a forward process where noise is gradually added to the training data, and a reverse process where noise is removed to restore the original data. By

learning the conditional probability density functions during the diffusion process, the model can eventually generate new samples—from complex data distributions such as high-resolution images. Inspired by diffusion in physics, the Diffusion Model operates by progressively simplifying the data’s structure and then reconstructing it. It has emerged as one of the most prominent paradigms in generative modeling, achieving excellent performance in computer vision tasks such as image generation, restoration, and transformation. Compared to GANs, Diffusion Models offer more stable training, produce high-quality images, and readily incorporate conditional inputs such as text or class labels. Furthermore, because the model’s outputs improve step by step during the reverse process, it provides advantages in transparency and interpretability by allowing analysis and control of intermediate stages.

Forward Process In the forward process of a Diffusion Model, Gaussian noise is progressively added to the original data x_0 , thereby destroying its underlying structure. This procedure is based on a Markov chain, and it is designed so that the data eventually approaches a nearly pure Gaussian distribution $N(\mathbf{0}, \mathbf{I})$. The method of noise addition at each time step $t \in 1, 2, \dots, T$ can be expressed as in Eq. 1 [15]:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; x_{t-1}\sqrt{1-\beta_t}, \beta_t \mathbf{I}\right) \quad (1)$$

The β_t is a variable representing the noise schedule at time step t , controlling the variance of the noise added at each step. Typically, β_t is set to gradually increase with t , so that as t approaches the final step T , the x_t almost completely loses the information of the original data and converges to pure noise. By injecting noise in this gradual manner, the model can learn the conditional distribution necessary for restoring the original data in the reverse process, even though the complex structure of the data is progressively diluted.

In the forward process, each stage’s output depends only on the previous output x_{t-1} , reflecting a Markov property. This chain structure simplifies complex transformations, providing a basis for efficient model training. In particular, there exists a closed-form interpretation for direct sampling x_t from x_0 , allowing the data at any time t to be computed all at once. This characteristic enhances the efficiency of training, and the probability distribution defined by the forward process serves as a crucial foundation for the reverse process.

Reverse Process In the Reverse Process, starting from data x_T that is saturated with noise, the model leverages its learned

parameters θ to progressively remove noise and restore a sample x_0 that closely resembles the original data. Each step in the reverse process can be considered a Markov process, in which the next state is predicted solely from the previous step’s data. Typically, it is modeled by a conditional probability of the form shown in Eq. 2:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right) \quad (2)$$

The $\mu_\theta(x_t, t)$ denotes the predicted mean, and $\Sigma_\theta(x_t, t)$ indicates the degree of noise removal, typically represented by a covariance matrix. In some implementations, this is fixed to a constant value such as $\beta_t \mathbf{I}$. During training, the model learns to predict the noise component ϵ from x_t , usually by employing a noise prediction network $\epsilon_\theta(x_t, t)$. A common loss function involves minimizing the difference between the predicted noise and the actual added noise, which is equivalent to optimizing the variational lower bound. Specifically, by training the model to minimize the Mean Squared Error (MSE) loss $L(\theta) = \mathbb{E}_t, x_0, \epsilon [|\epsilon - \epsilon_\theta(x_t, t)|^2]$, the model is guided to perform accurate noise removal at each step.

In the generation phase using the learned Reverse Process, one begins with a pure noise sample $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively applies the conditional distributions $p_\theta(x_{t-1} | x_t)$ predicted by the model. By traversing the path $x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$, new data is generated. At the end of this process, x_0 becomes a new sample with characteristics similar to the original training data. This step-by-step noise removal procedure is the key mechanism that enables Diffusion Models to generate high-resolution images, showcasing particular strength in effectively learning complex data distributions and capturing fine details.

2.2. SDXL

Stable Diffusion, a diffusion-based generative model epitomized by the original Stable Diffusion, has recently garnered significant attention in the field of image generation. However, earlier versions of Stable Diffusion faced quality degradation problems when generating high-resolution images, especially in terms of fine detail and overall consistency. To address these challenges, Stable Diffusion XL (SDXL) was proposed [16]. SDXL extends the original Stable Diffusion to substantially improve the quality and resolution of generated images and introduces a two-stage process that enables more refined image creation. Specifically, it adopts a two-step approach involving a Base model and a Refiner model, which generate and refine

images in succession. By allowing these two sub-models to perform different roles but cooperate with each other, SDXL is able to produce high-resolution images rich in detail without additional post-processing or upscaling.

- The Base model rapidly creates the overall outline and main composition of an image based on the user's prompt (text or other conditions). In this stage, a rough image is generated in the latent space, focusing on establishing the general structure and basic forms of the image. The Base model's key characteristic is to quickly capture the broad outline while leaving intricate details to the next stage. It also efficiently handles high-dimensional image data by converting them into a lower-dimensional latent representation. In addition, the user's prompt is used as a conditional input to reflect the desired style or composition.
- The Refiner model takes the latent image from the Base model and applies further refinement. It performs noise removal and detail enhancement on the Base model's output, making precise adjustments to texture, color, and subtle structures, ultimately producing the final high-resolution image. The Refiner model's primary characteristic lies in its ability to complement the finer points that may have been overlooked in the Base model's output, thereby enhancing realism and removing unnecessary noise for a clean and sharp result. Notably, it is designed to produce a high-resolution final image without a separate upscaling step.

Through the collaborative effort of these two models, SDXL effectively resolves the quality deterioration issues commonly seen in previous models when generating high-resolution images, enabling high-quality results in a single pass. In summary, SDXL overcomes the limitations of the original Stable Diffusion model and provides the following key advantages:

- **High-Resolution Image Generation:** By leveraging the synergy of the Base and Refiner models, SDXL can directly generate high-resolution images without separate post-processing or upscaling especially beneficial for tasks where detailed imagery is critical, such as accurately representing fine textures and patterns in fashion apparel as targeted in this study.
- **Extensive Parameter Utilization:** SDXL employs a very large number of internal parameters, enabling it to learn the intricate characteristics and details of complex images and thereby produce a wide range of styles and compositions reliably.
- **Efficient Computational Structure:** Despite its large number of parameters, the model's optimized two-stage design pre-

vents significant slowdowns or excessive resource consumption, allowing for fast generation speeds while maintaining high-quality outputs.

- **Versatile Applications:** SDXL can be used for various vision tasks such as text-based image generation, image restoration, and super-resolution transformation. In particular, it excels at producing images that accurately reflect a user's intended style or composition.

Ultimately, SDXL can be viewed as an extended version of Stable Diffusion capable of meticulously creating both the overall structure and fine details of images through a two-stage generation process. Thanks to its vast number of parameters and efficient computational design, it swiftly generates high-resolution images and is readily adaptable to various applications. Future research directions aim to further optimize the SDXL architecture and expand its capabilities to accommodate even more.

2.3. Flux

The Flux model is a diffusion model that adopts a distinctive approach compared to conventional image generation methods, utilizing a Diffusion Transformer architecture that combines the diffusion process with a Transformer network. This fusion of two technologies is the key factor enabling Flux to generate high-quality images while maintaining strong consistency with text conditions. As mentioned previously, the diffusion process serves as the fundamental mechanism of image generation: it starts with random noise and progressively removes the noise to form the image. During this process, the model gradually restores details such as texture, color, and shape, making it particularly effective for high-resolution images or intricate details. Simply put, it begins with blurred noise and incrementally refines it into a clearer image.

Meanwhile, the Transformer architecture (widely used in natural language processing) has strengths in deeply understanding the semantics and context of the input text. In the Flux model, this Transformer is employed to analyze text information and incorporate the extracted meaning into the image generation process. For instance, if given the sentence "a forest beneath a crimson sunset", the Transformer module infers visual features from the phrases "crimson sunset" and "forest", then encodes them into a vector representation. The Diffusion generation module then accepts this vector as a condition and goes through a noise-removal process, ultimately producing an image of the described scene.

- **Text Analysis:** The user-provided text prompt is encoded by the Transformer, which interprets its meaning and converts it into a vector representation.
- **Noise-Based Image Generation:** The Diffusion process begins with random noise and gradually forms the outlines and details of the image, guided by the text meaning provided by the Transformer.
- **Final Image Output:** After multiple steps of noise removal, the final high-quality image that corresponds to the text prompt is produced.

Through this procedure, Flux not only generates images but also precisely aligns visual elements with the semantics contained in the text, resulting in highly refined outputs. Leveraging extensive training data and a complex network structure, the Flux model can create extremely detailed and sophisticated images, preserving subtle details such as texture, color, and structure, while faithfully reflecting the user's intent. Moreover, thanks to the Transformer's powerful text comprehension capabilities, it can accurately map the content in the prompt to the elements of the image, thus achieving strong text-image consistency. For example, given a prompt like "a dragon floating above a medieval castle", Flux can generate a coherent image that captures both the features of a medieval castle and the appearance of a dragon.

2.4. Image Prompt (IP) Adapter

Text-to-Image models that generate images based on user-provided text prompts have been expanding rapidly in tandem with advances in AI technology. By automatically creating visual content from user-submitted sentences, these models serve as innovative tools in fields such as design, art, and entertainment. However, relying solely on text has limitations in accurately conveying complex visual details or specific styles. To overcome these challenges, methods that incorporate an additional Image Prompt (IP) have been explored, one of which is the IP Adapter.

IP Adapter is an extension module designed to effectively integrate image prompt functionality into a pre-trained Text-to-Image diffusion model. It acts as an add-on to the existing Stable Diffusion model, allowing users to input an additional image prompt without modifying the original architecture, thus enabling prompt-based image control without retraining the entire model. This makes the process easy to experiment with and quick to verify. By providing both a text prompt and an image prompt, one can generate higher-quality images.

IP Adapter operates by introducing an additional cross-attention layer into the existing diffusion model. This allows for specialized processing of text features by the original cross-attention layers and image features by the new ones, ensuring both modalities effectively guide the generation. Specifically, the original cross-attention layer in the UNet within the diffusion model continues to process text features, while IP Adapter adds a new cross-attention layer dedicated to image features. This new layer injects visual attributes (color, texture, shape, etc.) extracted from the image prompt into the model. During training, only the parameters of this newly added cross-attention layer are updated, while the original UNet weights remain fixed. As a result, the knowledge of the pre-trained model is preserved, and the additional functionality is learned efficiently without needing to retrain the entire model from scratch. Thanks to this attention decoupling approach, text information and image information do not interfere with each other and are both accurately reflected in the output. Moreover, one can easily add image prompt capabilities with minimal changes to the existing diffusion model architecture.

By leveraging IP Adapter, users can supplement the limited visual information provided by text alone with an image prompt, enabling the creation of more detailed and realistic images. For example, if a user inputs "red rose" as a text prompt and provides a real photo of a red rose as an image prompt, the model can merge the textual meaning with the visual features to generate an even more lifelike, intricate rose. In this way, the IP Adapter merges both text and image inputs to help users produce images with complex attributes and details. Moreover, since only the new cross-attention layer needs to be trained, there is no need to retrain the entire model, making it efficient in terms of computing resources. This approach not only facilitates rapid model expansion and easy application across a wide range of use cases but also retains the structure of the pre-trained diffusion model offering significant advantages in versatility.

2.5. Large Language Model

A Large Language Model (LLM) is an AI system based on the Transformer architecture, employing billions to hundreds of billions of parameters to learn linguistic grammar, semantics, and context from massive text corpora. For example, OpenAI's GPT-3, which has 175 billion parameters, is a representative LLM known for its remarkable language-processing capabilities. It uses a self-attention mechanism to precisely analyze

relationships among words in lengthy contexts. Through pre-training, LLMs acquire broad linguistic knowledge via self-supervised learning and then undergo fine-tuning to achieve specialized performance for specific tasks. Consequently, they can perform high-level natural language processing functions such as comprehending long contexts, generating fluent sentences, machine translation, and text summarization. These capabilities enable applications in chatbots, content creation, programming assistance, education, healthcare, and more.

However, LLMs also face several challenges. Their enormous number of parameters requires vast computational resources for training and inference, and biases intrinsic in the training data can influence model outputs. Additionally, these models sometimes exhibit hallucination, generating plausible-sounding content that does not actually exist in the context. To address such issues, ongoing research is needed to ensure ethical and efficient approaches to LLM usage.

Llama Llama is an LLM developed by Meta that boasts outstanding performance in the field of Natural Language Processing (NLP) [17,18]. Trained on billions of text data points, this model is capable of performing a wide range of language tasks such as text generation, machine translation, context-based question answering, and summarization. The Transformer architecture underlying Llama is particularly strong at handling long contexts, enabling it to effectively link and understand information spanning hundreds or even thousands of tokens within a document. For example, it can maintain consistency in its responses during extended conversations or while processing complex documents without losing track of earlier content. Another major advantage of Llama is its open-source availability. Researchers and developers worldwide can freely download the model, analyze its source code, and modify it according to their needs. This openness lowers the barriers to entry for large language model research and is considered a key factor in fostering collaboration between academia and industry.

Llama is offered in various parameter sizes, including Llama-7B (7 billion parameters), Llama-13B (13 billion), and Llama-70B (70 billion). Each version is designed to strike a balance between resource consumption and performance—ideal for using lighter models in smaller research environments or deploying large-scale models on powerful computing resources. Furthermore, Llama’s performance can be maximized for specific domains through fine-tuning. For instance, medically specialized fine-tuning allows Llama to analyze research articles or patient records and provide accurate responses using domain-

specific terminology. Mixed Precision Training helps improve computational efficiency by reducing GPU memory usage, and Gradient Checkpointing enables large-scale model training. Thanks to these technical advantages, Llama maintains high performance even in resource-limited environments, demonstrating its practical utility across diverse applications.

2.6. Prompt Tuning

Prompt Tuning is a technique for improving an LLM’s output by adjusting the input prompt rather than retraining the model itself. This method involves making slight changes to the wording or structure of the prompt to enable the model to perform better on a specific task. By doing so, the model can more accurately grasp the context and intent behind the prompt and generate outputs aligned with that context. For example, instead of a simple request like “Summarize this sentence,” providing a more explicit prompt such as “Please explain the main points of the following sentence in two concise sentences” leads the model to produce a more precise summary.

A key aspect of Prompt Tuning is that it optimizes only the prompt without altering the model’s internal weights. This makes it an efficient approach that preserves existing learned knowledge while enhancing task-specific performance. In contrast, fine-tuning involves re-training the model’s weights on a particular dataset, which demands extensive computing resources and time. Since Prompt Tuning does not modify the model itself, it requires fewer resources and can be applied more rapidly. In this study, we employ Instruction Tuning and CoT as part of our Prompt Tuning strategy.

Instruction Tuning Instruction Tuning is one method for enhancing the performance of large-scale language models by combining the strengths of fine-tuning and in-context learning [19]. In this technique, the model is additionally trained on data in the form of instructions for various tasks, adjusting it so that it produces appropriate outputs in response to user-provided directives. In other words, it is a form of instruction-based fine-tuning that enables the model to understand explicit commands and respond accordingly. As a result, the model can quickly adapt to new tasks without requiring modifications to its architecture. For example, if the model is trained on an instructional sentence like “Rewrite this sentence in a positive tone,” it will subsequently be capable of reconstructing any given sentence into a positive tone when that instruction is provided.

The primary advantage of Instruction Tuning is that it enables the large, pre-trained model to be applied to a wide vari-

ety of tasks without additional retraining. Conventional fine-tuning involves further training the entire model on a new dataset for each task, requiring large amounts of data and extended training time. In contrast, a model that has undergone Instruction Tuning can immediately generate task-appropriate outputs by simply being given an instruction—no further training needed. For instance, if told “Summarize this text in three sentences,” the model can produce a summary in three sentences without additional training. Thanks to this trained ability to interpret directives, the model can swiftly handle new tasks and acquires general-purpose language generation capabilities across a broad range of user instructions. Consequently, Instruction Tuning is considered a useful technique for effectively deploying large language models in various applications, particularly when additional data or time resources are limited.

2.7. Chain of Thought

Chain of Thought (CoT) is a prompt-design method that guides a model to explicitly generate intermediate reasoning steps when solving complex problems [20,21]. In standard few-shot learning, a few examples are provided for the model to infer the pattern between inputs and outputs. However, in CoT, the logical train of thought required to arrive at the answer is included in the example solutions, helping the model learn the problem-solving process alongside the final answer. In other words, rather than presenting just the question and the final answer, one also includes a step-by-step explanation of why that answer is derived.

By doing so, the model goes beyond simple input-output mapping to learn the procedure of problem-solving, increasing the likelihood that it will arrive at the correct solution for more complex tasks. For instance, imagine a scenario involving math problems: instead of merely instructing “Solve the following equation,” one might provide a prompt like “First add the given numbers, then double the result, and finally compute the answer.” When the model uses these step-by-step instructions, it can imitate the intermediate calculation steps and produce a more systematic response—thus increasing the probability of arriving at a correct result. CoT has been reported to significantly improve model performance in tasks requiring complex reasoning, such as math problem solving or logic puzzles. Because the model not only learns the answer itself but also the rationale behind it, it can maintain logical consistency when faced with new problems, carrying out its reasoning in a coherent manner.

Both Instruction Tuning and the CoT method are powerful

tools for enhancing the capabilities of large language models, and each can individually yield notable improvements in performance. Furthermore, combining these two methods can produce a synergistic effect greater than that of applying them separately. In this study, we leverage the synergistic strengths of both methods. First, Instruction Tuning is used to extract key information from user prompts and expand it into more detailed instructions. Then, CoT prompting structures the step-by-step generation process for these detailed front and back descriptions. This approach plays a crucial role in automatically generating detailed and consistent prompts for both the front and back sides of clothing images, ultimately enabling the construction of a highly coherent double-sided image dataset.

3. Methods

3.1. Pipeline overview

The overall pipeline used in this study is shown in Fig. 1. The prompt received from the user can be as simple as “sky-blue shirt” which is first processed by the LLM-based prompt expansion module. We use the large-scale open-source Llama3-Instruct language model in this module for ease of reproducibility. In this module, instruction tuning systematically expands upon the essential elements of the clothing description such as color, material, and design details. Subsequently, CoT ensures logical consistency for these attributes between the generated front and back prompts. The two resulting detailed text prompts are then passed on as inputs to the image generation modules.

First, the front prompt is fed into the FLUX diffusion model to generate a front-view image of the garment. Next, the back prompt is provided to the SDXL diffusion model, which uses the IP Adapter to incorporate the front-view image. By referring to both the text and the front image, SDXL is able to maintain design consistency between the front and back while generating the rear view. This series of steps is fully automated, so only a single text prompt from the user is required to produce a paired set of front and back clothing images.

The key aspect of this pipeline lies in maximizing consistency by leveraging both textual and visual information. The LLM-generated front and back prompts describe the same garment while reflecting the differences between the two views. During the diffusion step, the visual features extracted from the front image are reused in the back-image generation. In the following section, we describe each step of prompt creation and

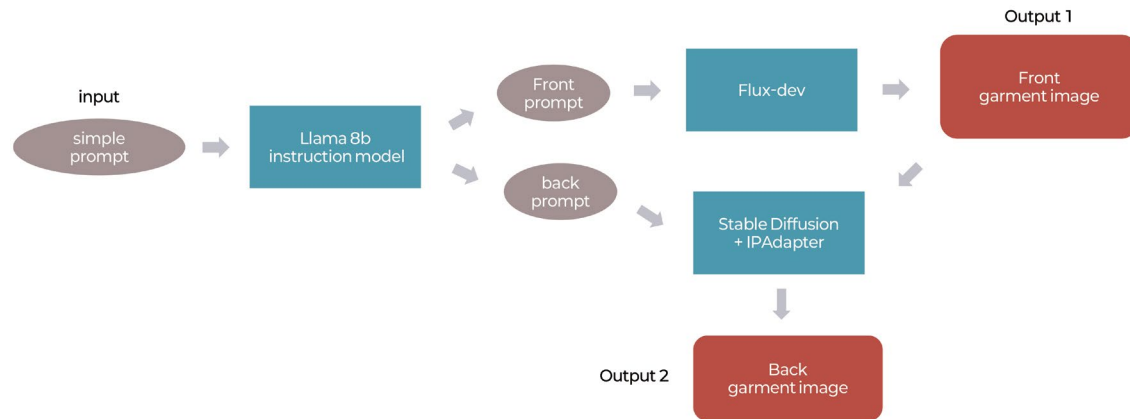


Fig. 1. Overview of the pipeline for generating front and back clothing images.

image generation in detail.

A brief text description is provided as input, which the LLM uses to generate detailed prompts for both the front and back views. These prompts are then fed into a diffusion model, yielding the correlated front image (a) and back image (b). By leveraging the IP-Adapter module, the features of the front image are utilized in creating the back image, ensuring that the design and attributes of the two outputs remain consistent.

3.2. Prompt generation using an LLM (Instruction Tuning + CoT)

The user's natural language prompt is first processed by a prompt expansion module. In this study, we employed a fine-tuned InstructGPT series model based on Llama3 to handle this task. We controlled the model's output through careful prompt design so that it could produce descriptive fashion-related content without additional fine-tuning.

Prompt expansion takes place in two steps. First, using Instruction Tuning, a one-line description input by the user is transformed into detailed instructions for the front and back views. For example, if the user enters "sky-blue shirt," the model interprets this input and divides it into a Front Prompt (e.g., "Generate a front image of the shirt") and a Back Prompt (e.g., "Generate a back image of the same shirt"). Each prompt contains detailed references to color (sky-blue), material (e.g., cotton or silk), and design details (e.g., embroidery patterns, button layouts, hem shapes). These elements are automatically incorporated based on a predefined list of fashion attributes and general knowledge; the LLM uses its large-scale knowledge to fill in any gaps.

Next, the CoT guidance technique is applied to ensure logical

consistency between the front and back prompts. Through CoT, the LLM methodically reviews the design elements mentioned in the front prompt so they carry over to the back prompt. For instance, if the front prompt states, "There is a white embroidered pattern along the front placket," the CoT process for the back prompt might include a phrase like, "There is a matching white embroidered pattern along the back yoke." By maintaining consistent context in its output, the model ultimately produces front and back prompts that describe the same garment as a coherent pair.

The specific two-stage image generation pipeline in this study involves using the FLUX model for the initial front view generation, followed by the SDXL model combined with an IP-Adapter to generate the corresponding back view, conditioned on the front image. This configuration was deliberately chosen based on the distinct capabilities and current limitations of these models. SDXL, while powerful, often includes human figures in generations even from garment-only prompts, making it less ideal for producing clean initial reference images. Conversely, FLUX tends to generate garment-only images more reliably from text but currently lacks robust, official support for the direct image-to-image conditioning (via IP-Adapter or similar) needed for the back view's consistency with the front. Therefore, our approach leverages FLUX's strength for a clean front image, then employs the superior image conditioning capabilities of SDXL with the IP-Adapter to ensure high consistency and detail in the subsequent back view generation, presenting the most effective pathway to achieve our objectives.

In summary, the prompt expansion module operates in the following sequence: [User → Input] → [instruction tuning] → [Draft Descriptions for Front and Back] → [CoT Verification]

→ [Final Front and Back Prompts]. The resulting Front Prompt and Back Prompt each contain sufficient information to render the garment from their respective angles while ensuring that the core attributes of the clothing are consistently described between the two prompts.

3.3. Front image generation (FLUX Model)

The FLUX model takes the previously generated Front Prompt as input to synthesize the front-view image of the garment. The FLUX model was chosen for front image generation because, compared to some existing Stable Diffusion-based models, it tends to preserve richer detail and adhere more strongly to the input prompt. This strong prompt adherence is particularly useful for effectively validating how our CoT-based textual descriptions influence the final image.

Because the Front Prompt contains highly specific visual details about the clothing, FLUX uses this information to materialize the user’s intended image. For instance, if the front prompt states, “A sky-blue silk blouse with a subtle sheen, featuring intricate white embroidery along the front placket,” FLUX converts this sentence into vector embeddings and processes them through its diffusion transformer network, step by step transforming noisy images into refined garment images. In each step of this noise-removal process, the transformer interprets the textual meaning (e.g., color = sky-blue, material = silk, detail = embroidery) and incorporates these characteristics into the evolving image.

Thanks to FLUX’s robust text-to-image mapping, all elements explicitly mentioned in the prompt are faithfully rendered in the front image. In particular, because the CoT process produces prompts with contextual descriptions rather than simple phrases, FLUX can balance the overall shape of the garment with its fine-grained details. The resulting front image is output at a resolution of 512×512 pixels, then serves as a visual reference for the subsequent back-image generation phase. If FLUX generates multiple samples, we select the result best matching the prompt, such as the image that clearly displays the described embroidery, for the next stage. The front image thus obtained functions as a kind of intermediate output in the pipeline illustrated in Fig. 2.

3.4. Back image generation (SDXL + IP Adapter)

The back-view image is generated by feeding the Back Prompt into the SDXL model. While SDXL on its own has powerful text-to-image capabilities and can depict the rear view

of the garment solely from that prompt, our method further enhances front-back consistency by incorporating additional visual information from the front image. This is facilitated by the IP Adapter module. The IP Adapter receives the front image (produced in the previous step by FLUX) and converts it into an image embedding, which is then integrated into SDXL’s diffusion process.

By using both the Back Prompt and the feature representation of the front image, SDXL can synthesize the back view, simultaneously considering textual information (such as the garment’s shape or back-specific details) and image-based information like color and texture from the front. Consequently, visual elements from the front image are carried over to the back view. For instance, if a particular logo is present on the front, the back view might include a subtly rendered or appropriately placed version of the same logo. Similarly, textures displayed on the front are reflected in the back. This process leads to a significant increase in visual consistency between the two images. Fig. 2(b) provides an example of a back view generated with reference to the front image in panel (a), demonstrating that color and fabric texture match perfectly. The resolution of the SDXL output is 512×512 pixels. If needed, minor upscaling and noise reduction can be applied in post-processing to produce the final image.

The greatest advantage of the SDXL + IP Adapter step is the preservation of context for the same garment. While the front and back images stand on their own, when viewed as a pair, they exhibit continuity. Without the IP Adapter, the back view generated from the Back Prompt alone could be largely unrelated to the front. However, with the IP Adapter, the design language evident in the front image is extended to the back view. This ensures that users obtain a consistent set of clothing images, with enough reliability to be directly applicable to datasets such as VTON.

4. Experiment

4.1. Experimental environment and data configuration

To evaluate the performance of the proposed method, we conducted generation experiments. The implementation was done in Python 3.10 using the PyTorch framework, and all generation steps were performed on a workstation equipped with an NVIDIA RTX 3090 GPU with 24 GB of VRAM. This setup handles the large-scale computations required by diffusion

models, enabling high-quality image outputs.

First, we verified the operation of our prompt-expansion module driven by an LLM, and then generated front-and-back clothing images using a combined FLUX and SDXL + IP Adapter model. As an experimental dataset, we compiled our own set of 50 short text descriptions for various clothing items. Specifically, we collected brief prompts that covered a wide range of fashion categories (such as “red hoodie”, “ivory shirt”, and “brown leather jacket”) and generated a front-back image pair for each prompt via the pipeline described in this paper.

For comparison, we also obtained baseline results using the conventional approach: we entered the same short sentences directly into the SDXL model to independently generate front and back images. No IP Adapter was employed in the baseline, so the two images were created solely based on text conditions, without any direct correlation to each other. This allows for a fair comparison between existing text-to-image generation methods and our proposed approach.

In the case of Fig. 2, the user enters a simple prompt, from which the llama3 model (leveraging CoT and Instruction tuning techniques) generates detailed front and back prompts. Through subsequent steps, front and back images are then created. Examples of other types of clothing are described in Appendix A. Below is a concise example of the prompt for the first case, sky-blue shirt.

- Front Prompt: A sky-blue, lightweight, silk-blend shirt with a classic collar, slim-fit silhouette, and intricate white embroidery along the front placket. The fabric has a subtle sheen

and a soft drape. The shirt features a single-button closure and a slightly curved hem.

- Back Prompt: The same sky-blue, lightweight, silk-blend shirt with a classic collar, slim-fit silhouette, and intricate white embroidery along the back yoke. The fabric has a subtle sheen and a soft drape. The shirt features a single-button closure and a slightly curved hem, with a subtle seam running along the center back.

4.2. Evaluation metrics

To quantitatively assess the consistency, quality, and diversity of the front-back clothing images generated by our proposed two-stage diffusion pipeline, we have adopted the following three key metrics. Each metric evaluates different critical aspects of the generated images, providing a comprehensive understanding of the proposed methodology’s performance.

Learned Perceptual Image Patch Similarity (LPIPS) Learned Perceptual Image Patch Similarity (LPIPS) is a widely used metric for measuring the perceptual similarity between two images [22]. Unlike traditional metrics such as Mean Squared Error (MSE), which calculates pixel-wise differences, LPIPS is designed to align more closely with human visual perception of similarity. It achieves this by computing the distance between feature vectors extracted from intermediate layers of pre-trained deep neural networks. A core objective of this study is to ensure high visual consistency in key attributes (design, color, material) between the front and back views generated from a simple user prompt. LPIPS is therefore highly effective for quantitatively



Fig. 2. Generative AI image of a sky-blue shirt. (a) is the front image and (b) is the back image.

evaluating this visual consistency between the front and back image pairs. A lower LPIPS value indicates a higher perceptual similarity between the two images, signifying that the generated front and back views are consistent as different perspectives of the same clothing item.

Fréchet Inception Distance (FID) Fréchet Inception Distance (FID) is one of the standard metrics to evaluate the overall quality and diversity of images produced by generative models [23]. FID measures the fr chet distance between the feature distributions of real images and generated images, where features are extracted using a pre-trained Inception v3 model. This distance considers both the mean and covariance of the feature activations. A lower FID score suggests that the distribution of generated images is statistically similar to that of real images, indicating greater realism and diversity in the generated outputs.

In this study, FID was chosen to comprehensively assess how closely the generated clothing images resemble actual clothing items and to gauge their overall quality. This helps in determining the practical utility of the generated clothing image dataset.

Precision and Recall To provide a more nuanced understanding of the performance of the generative model, we also consider Precision and Recall for the distributions of generated images [24,25]. These two metrics are valuable for separately evaluating the quality and diversity of the generated images.

The Precision metric measures how well the generated images fall within the distribution of real images. In other words, it evaluates the fidelity or realism of the generated images. A high precision value indicates that the generated images are of high quality and are likely to be mistaken for real images, which is crucial for the credibility of generated content in applications like VTON.

The Recall metric assesses how well the generated images cover the diversity of modes present in the real image distribution.

A high recall value suggests that the model has learned various features from the real data, such as different styles, shapes, colors, and can produce a wide range of images.

This research aims not only to generate consistent front-back clothing images, but also to contribute to the construction of diverse and high-quality image datasets for various applications. Therefore, Precision and Recall play a vital role in quantitatively evaluating the realism (Precision) and variety (Recall) of the generated clothing images. These metrics thereby provide a multi-faceted validation of the proposed pipeline’s utility.

5. Results

5.1. Qualitative evaluation

Fig. 2 presents the front and back views of a sky-blue shirt generated by our proposed pipeline. Panel (a) shows the front view created by the FLUX model using the expanded Front Prompt, and panel (b) is the back view generated by the SDXL model after incorporating visual information extracted from the front image via the IP Adapter. Both images share the same prompt and visual details, maintaining a consistency that makes them appear as if they were taken from different angles of the same shirt.

As another example, Appendix Fig. 1 in Appendix A shows the results for a red hoodie. Panel (a) depicts the front view generated by our proposed pipeline, while panel (b) shows the corresponding back view. Both images clearly express the deep red fabric texture and the same logo design on the front and back, providing consistent visual information akin to photographs of a single hoodie captured from different perspectives. By effectively applying the visual information from the front image to the generation of the back image, the IP Adapter module resolves inconsistencies often encountered in text-only image generation, such as having a logo on the front but missing or differently placed on the back.

Appendix A further illustrates front and back images of various types of clothing generated through our proposed pipeline. In all cases, visual features from the front images, such as the number of buttons, lace patterns, and glossiness are consistently reflected in the back images as well. This outcome would be difficult to achieve with simple text-based generation methods, suggesting that our integrated LLM + Diffusion approach opens new possibilities for clothing image generation.

Nonetheless, as shown in Appendix B, there are examples of incorrectly generated images. Appendix Figs. 11 and 12 display clothing being worn by a person, which deviates from our goal of showing the clothing item itself. In addition, Appendix Fig. 13 shows an error where the back image either appears the same as the front or is missing altogether, leaving only the correct front image. These errors highlight areas needing improvement, but they also indicate the potential usefulness of this current approach for acquiring large-scale training data in a generative fashion AI without the need for extensive photo shoots. Reasons for failure and future improvements are detailed in Appendix B.

In summary, our proposed pipeline (comprising Prompt Expansion, FLUX, and SDXL with IP Adapter) demonstrates

the ability to generate consistent, high-quality pairs of clothing images from simple text input. Such data could be used in the future fashion AI domain to obtain vast amounts of training data without manual photography. While the current method is limited to front and back views, it theoretically holds the potential to expand to multi-view image generation (e.g., side or 45-degree angles). In the next chapter, we present the conclusions of this study and discuss future research directions, including these potential expansions.

5.2. Quantitative evaluation

To quantitatively assess the performance of the two-stage diffusion pipeline proposed in this study, an analysis was conducted on the generated front-back clothing image pairs using the key evaluation metrics defined in Section 4.2: LPIPS, FID, and distribution-based Precision and Recall. The quantitative results for representative generation outcomes, including both successful and failure cases, are summarized in Table 1. This table includes examples presented in Fig. 2 and Appendix A, as well as failure cases from Appendix B, with their respective scores for each metric.

As shown in Table 1, most successfully generated clothing items (e.g., “Sky-blue shirt,” “Red hoodie,” “Brown trench coat”) achieved excellent scores across the newly adopted quantitative metrics. Specifically, for the ‘Sky-blue shirt’ shown in Fig. 2, the LPIPS between the generated front and back images was a low 0.12. This low LPIPS value indicates high perceptual consistency. It signifies that the delicate white

embroidery pattern, the subtle sheen and texture of the fabric, and the overall silhouette of the shirt depicted in the front view were all maintained with high visual fidelity in the back view. The FID for this image was 40. This low score indicates that the generated shirt image achieved high realism and overall quality, characterized by natural color tones and sharp details without artificial artifacts. Furthermore, the ‘Sky-blue shirt’ obtained a high Precision score of 0.85, as it faithfully implemented key visual attributes specified in the detailed prompt, such as ‘sky-blue,’ ‘lightweight, silk-blend,’ ‘classic collar,’ and ‘intricate white embroidery along the front placket.’ Concurrently, it scored well on Recall with 0.75, effectively capturing rich visual information like the soft draping characteristic of silk, subtle gloss, and the three-dimensional feel of the embroidery.

Another successful example, the ‘Brown trench coat’ shown in Appendix Fig. 10, also demonstrated outstanding results. Complex structural elements such as the double-breasted design, belt, and epaulets on the front view corresponded accurately with their respective parts on the back view, maintaining a consistent style, which resulted in a very low LPIPS of 0.11. The realistic rendering of a leather-like texture and effective light/shadow play contributed to an excellent FID of 35. Its Precision reached 0.88 due to the accurate reflection of prompt details like ‘classic double-breasted design,’ ‘water-repellent,’ and ‘fitted silhouette,’ while its Recall was 0.82, successfully capturing various visual features including complex design elements and fabric texture.

Conversely, some failure cases detailed in Appendix B exhibited poorer performance on specific metrics, as anticipated. For instance, in the ‘Jeans’ shown in Appendix Fig. 11 and ‘Blue hoodie’ shown in Appendix Fig. 12 examples, the garments were generated being worn by a virtual model. Although the color and main patterns of the garments themselves might have maintained some consistency between the front and back views, their LPIPS scores were 0.25 and 0.22, respectively. These values were slightly higher than those of the successful cases. This suggests that the presence of non-clothing elements, such as the model and background, could interfere with the LPIPS assessment of pure garment-to-garment consistency. Furthermore, the inclusion of these unintended elements negatively impacted the overall image quality and its distributional similarity to ‘garment-only’ images, leading to higher FID values of 90 and 85, respectively. As the generated images did not accurately represent the targeted ‘garment-only’ characteristics, their Precision scores were relatively low at 0.65 and 0.68.

Table 1. Quantitative evaluation results for generated front-back clothing image pairs

Item description	LPIPS	FID	Precision	Recall
Sky-blue shirt	0.12	40	0.85	0.75
Red hoodie	0.15	45	0.82	0.70
Ivory blouse	0.14	42	0.83	0.72
Purple silk blouse	0.13	40	0.84	0.78
Black tank top	0.18	50	0.80	0.65
Ivory cotton pants	0.16	48	0.81	0.68
Brown leather jacket	0.12	38	0.86	0.80
Gray sweatshirts	0.15	45	0.82	0.70
Yellow sweatshirts	0.14	43	0.83	0.71
Pink cardigan	0.13	40	0.84	0.75
Brown trench coat	0.11	35	0.88	0.82
Jeans	0.25	90	0.65	0.60
Blue hoodie	0.22	85	0.68	0.62
Mint windbreaker	0.90	180	0.50	0.40

The most notable failure was the ‘Mint windbreaker’ shown in Appendix Fig. 14, where a front view was erroneously generated in response to a back view request. Consequently, a meaningful comparison between an intended front and back pair was impossible, leading to a very high LPIPS of 0.90, clearly indicating a failure in perceptual consistency. This manifest generation error also severely degraded image realism, reflected in a very high FID of 180. As the output failed to represent any features of the intended back view, both Precision 0.50 and Recall 0.40 were very low. This error is attributed to a flaw in the LLM prompt generation stage, where an explicit front-view feature “zip-up closure” was mistakenly included in the back-view prompt as described in Appendix B.3, likely confusing the image generation model into regenerating a front view.

In conclusion, the quantitative evaluation using LPIPS, FID, Precision, and Recall metrics demonstrates that the proposed two-stage diffusion pipeline can effectively generate consistent and realistic front-back images for various clothing items, as evidenced by the successful cases. Concurrently, the failure cases clearly highlight specific areas for future improvement, corroborating the qualitative findings discussed earlier. These quantitative results contribute to a more objective understanding of the proposed methodology’s effectiveness and limitations.

6. Conclusion

In this study, we propose a novel methodology for generating consistent front-and-back clothing images by applying CoT and Instruction Tuning techniques. This approach represents an effort to overcome the limitations of existing datasets in the Virtual Try-On (VTON) field. It demonstrates that high-quality pairs of clothing images can be obtained with only simple text input. Specifically, through LLM-based prompt expansion, we generate refined and consistent front and back prompts from the user’s input. These prompts are fed into the FLUX and SDXL diffusion models, with the latter incorporating the IP Adapter module. Through this process, we successfully synthesize clothing images where the designs on the front and back match seamlessly. Experimental results show superior consistency and accuracy, and various examples confirm the practical applicability of the method. Notably, the ability to acquire multi-angle images of desired clothing without additional photo shoots or complex 3D modeling suggests a new paradigm in constructing fashion datasets.

However, this study also has some limitations. First, the current pipeline generates clothing on a solid-colored background only. This makes it difficult to directly apply to images of people actually wearing the garments or those with complex backgrounds. Second, we have focused on just two angles (front and back) without addressing side views or intermediate angles like 45° perspectives. Third, because the prompt expansion process relies entirely on the LLM, there is a risk that unintended details may appear if the LLM applies incorrect assumptions about the garment. In some cases, the LLM might misunderstand the prompt. For instance, it sometimes generated a front-view image when the back view was requested. Finally, using large generative models such as FLUX and SDXL means that generating a single image can take several seconds. This can be time-consuming when producing large volumes of data.

To illustrate some of these limitations more concretely, we now discuss specific failure cases observed during our experiments. While the pipeline proposed in this study is effective for generating consistent front and back clothing images, unexpected results were observed in some cases.

We outline the following directions for future research to overcome these limitations:

- **Extension to Various Human Silhouettes and Poses:** Expand the proposed method to generate front and back views (and potentially other angles) of a person actually wearing the same garment in different poses. For instance, if provided with only a single front-facing photo of a person and the type of clothing, the technology could generate the back view. This would make a more realistic contribution to VTON systems.
- **Improving Efficiency and Lightweight Performance:** Enhancing real-time responsiveness by streamlining and optimizing the generation process is another goal. This might involve optimizing the pipeline from LLM prompt creation to diffusion model output or minimizing latency, facilitating interactive fashion design tools, or real-time online fitting services in the future.
- **Ethical Considerations:** As AI-generated fashion images become more practically useful, ethical considerations are also crucial. Measures must be established to address copyright issues, prevent data misuse, and avoid consumer confusion arising from clothing images that do not actually exist.

Going forward, we plan to propose guidelines for the ethical use of AI generated fashion images and conduct research to ensure that the proposed approach is socially acceptable. In

summary, our research integrates prompt engineering with generative models to revolutionize clothing image generation. It demonstrates the potential to resolve the limited data problem in fashion and to create abundant virtual clothing datasets. With further technical refinements and practical implementations, the technique has the potential to be directly utilized in a variety of fashion industry applications, including design simulation, virtual fitting for e-commerce, and digital clothing catalogs.

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APPENDIX

A. Generative Images with Prompt Commands

A.1. A red hoodie

- Front Prompt: Generate an image of a red hoodie with a soft, fleecy fabric, featuring a minimalist design with a subtle texture pattern. The hoodie has a relaxed fit with a rounded neckline and long sleeves. The zipper is centered and has a small, discreet logo on the chest. The color is a deep, vibrant red with slight undertones of orange.
- Back Prompt: Generate an image of the same red hoodie from the back, showcasing the same soft, fleecy fabric and minimalist design with a subtle texture pattern. The hoodie's relaxed fit and rounded neckline remain consistent, as well as the centered zipper and small logo on the back. The color remains the same deep, vibrant red with slight undertones of orange.

A.2. An ivory blouse

- Front Prompt: Create an image of a white blouson with a fitted silhouette, made from a luxurious silk fabric with a subtle sheen. The garment features a high neckline, long sleeves, and a flowing A-line design with intricate embroidered floral patterns in a delicate shade of gold. The fabric drapes elegantly, showcasing the garment's craftsmanship.

- Back Prompt: Show an image of the same white blouse from the back, maintaining the same luxurious silk fabric, fitted silhouette, and flowing A-line design. The embroidered floral patterns in a delicate shade of gold are still prominent, and the high neckline and long sleeves are visible. The garment's craftsmanship and fabric quality are also showcased in this angle.

A.3. A purple silk blouse

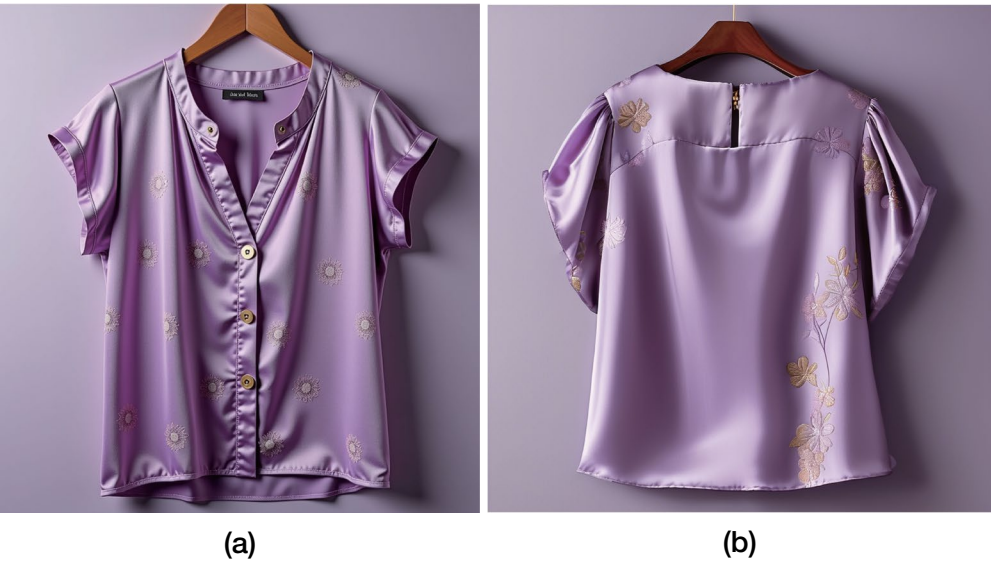
- Front Prompt: Generate an image of a purple silk blouse with a subtle sheen, featuring a delicate floral pattern in a lighter shade of purple. The blouse has a relaxed fit with a V-neckline and cap sleeves. The fabric appears smooth and lustrous, with a slight texture that catches the light. The design is intricate, with tiny details that add depth and dimensionality to the overall design. The blouse is the main focus, with no distractions or obstacles in the background.
- Back Prompt: Generate an image of the same purple silk blouse from the back, showcasing the same delicate floral pattern, subtle sheen, and smooth fabric texture. The V-neckline and cap sleeves remain prominent, and the relaxed fit remains consistent. The intricate design details are still visible, adding depth and dimensionality to the overall design. The blouse remains the main focus, with no distractions or obstacles in the background.



Appendix Fig. 1. Generative AI image of a red hoodie. (a) is the front image and (b) is the back image.



Appendix Fig. 2. Generative AI image of an ivory blouse. (a) is the front image and (b) is the back image.



Appendix Fig. 3. Generative AI image of a purple silk blouse. (a) is the front image and (b) is the back image.

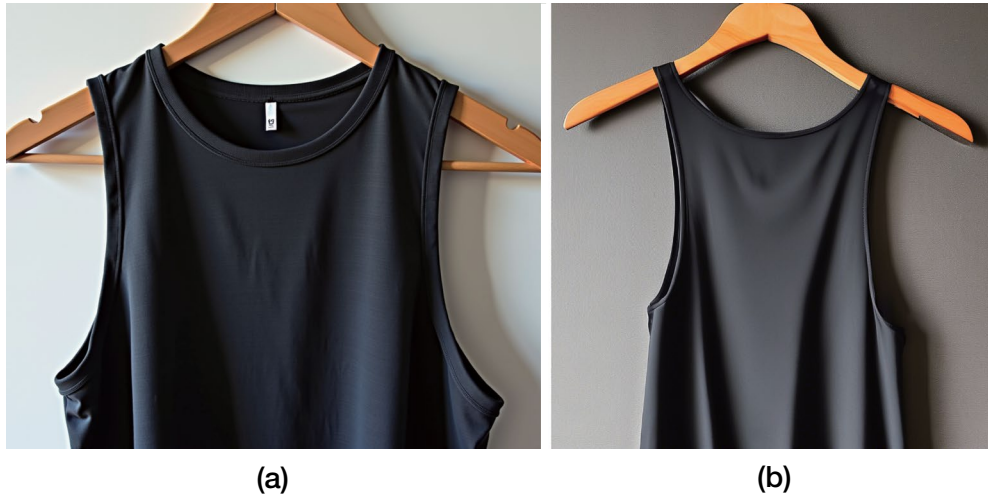
A.4. A black tank top

- Front Prompt: Generate an image of a black tank top made from soft, lightweight cotton fabric with a subtle sheen. The design features a minimalist, high-neckline silhouette with a fitted, curved shape. The fabric has a slight texture, giving it a slightly rumpled appearance. The black color is deep and rich, with a slight blue undertone.
- Back Prompt: Show the same black tank top from the back, maintaining the same fabric type, design details, and color as the front view. The high neckline and fitted silhouette should

be visible from the back, with the same subtle texture and slight rumpling effect. The black color should remain consistent, with the same blue undertone.

A.5. An ivory cotton pants

- Front Prompt: Generate an image of a pair of ivory cotton pants with a classic straight-leg design, featuring a subtle texture and a delicate button fly. The pants should have a relaxed fit with a slightly tapered leg, and a 7/8 length. The ivory color should be a soft, creamy white with a slight



Appendix Fig. 4. Generative AI image of a black tank top. (a) is the front image and (b) is the back image.



Appendix Fig. 5. Generative AI image of an ivory cotton pants. (a) is the front image and (b) is the back image.

sheen. Emphasize the fabric's natural texture and the pants' clean lines.

- **Back Prompt:** Generate an image of the same pair of ivory cotton pants from the back, showcasing the same classic straight-leg design, subtle texture, and delicate button fly. The pants should retain the same relaxed fit and 7/8 length as the front view. The ivory color should remain the same soft, creamy white with a slight sheen. Emphasize the pants' clean lines and the subtle texture of the fabric.

A.6. A brown leather jacket

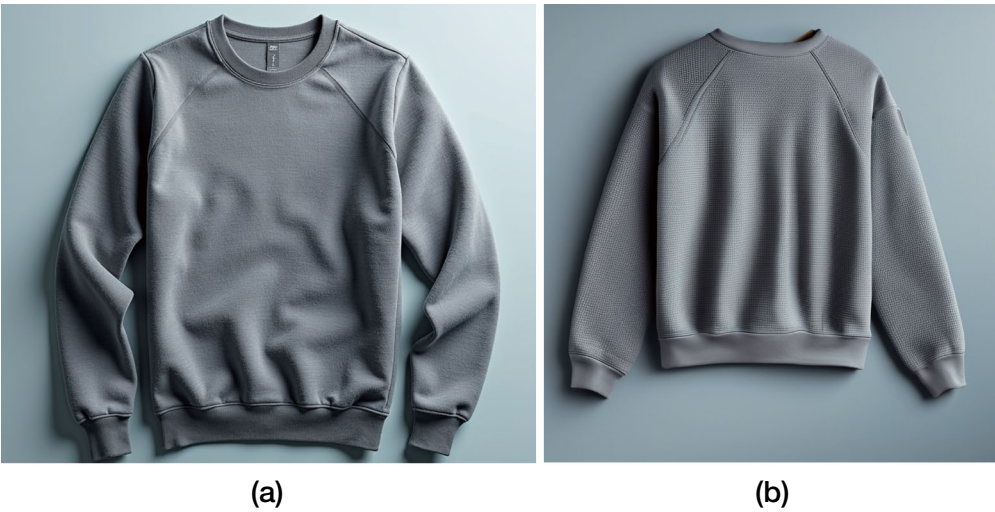
- **Front Prompt:** Generate an image of a brown leather jacket with a classic design, featuring a fitted silhouette, a zip-up

front closure, and a subtle sheen from the leather. The jacket should have a smooth, matte finish and a rich, earthy brown color. The leather should be high-quality, with a soft, supple texture and a slight grain pattern. The overall look should exude sophistication and elegance.

- **Back Prompt:** Show the same brown leather jacket from the back, maintaining the same design, color, and texture as the front view. The zip-up closure should be centered, and the seams should be clean and subtle. The leather should have the same smooth, matte finish and rich, earthy brown color as the front view. The overall appearance should be consistent with the front view, emphasizing the same high-quality leather and classic design.



Appendix Fig. 6. Generative AI image of a brown leather jacket. (a) is the front image and (b) is the back image.



Appendix Fig. 7. Generative AI image of a gray sweatshirts. (a) is the front image and (b) is the back image.

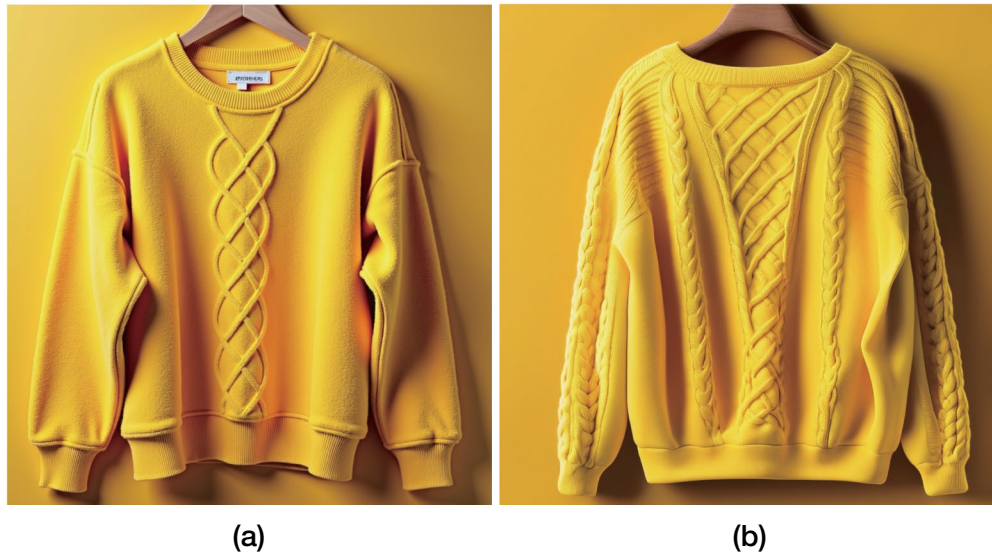
A.7. A gray sweatshirts

- Front Prompt: Generate an image of a grey sweatshirt with a soft, fleecy texture, featuring a minimalist design with a subtle grid pattern. The sweatshirt has a relaxed fit with a crew neck and long sleeves. The grey color is a medium to dark tone with a slight blue undertone. Emphasize the texture and fabric details.
- Back Prompt: Generate an image of the same grey sweatshirt from the back, showcasing the same soft, fleecy texture and minimalist design with a subtle grid pattern. The sweatshirt’s relaxed fit and long sleeves are still prominent, with the same medium to dark grey color tone and slight blue undertone.

Focus on the fabric details and texture consistency.

A.8. A yellow sweatshirts

- Front Prompt: Generate an image of a bright yellow sweatshirt with a soft, fleecy texture. The design features a subtle, cabled pattern and a crew neck. The sweatshirt has a relaxed fit with a slightly dropped shoulder. The fabric appears plush and warm to the touch. The overall style is casual and comfortable.
- Back Prompt: Show the same bright yellow sweatshirt from the back, maintaining the same soft, fleecy texture and subtle, cabled pattern. The crew neck remains the same, and the



Appendix Fig. 8. Generative AI image of a yellow sweatshirts. (a) is the front image and (b) is the back image.



Appendix Fig. 9. Generative AI image of a pink cardigan. (a) is the front image and (b) is the back image.

relaxed fit with a slightly dropped shoulder is consistent. The fabric appears plush and warm to the touch, with no changes in texture or appearance. The overall style remains casual and comfortable.

A.9. A pink cardigan

- **Front Prompt:** Generate an image of a pink cardigan with a soft, chunky knit fabric, featuring a subtle cable knit pattern on the front panel. The cardigan has a relaxed fit with a V-neckline and long sleeves. The pink color is a vibrant,

pastel shade with a slight sheen. The fabric has a slightly textured finish, giving it a cozy appearance.

- **Back Prompt:** Show the same pink cardigan from the back, maintaining the same fabric type, cable knit pattern, and relaxed fit. The V-neckline and long sleeves remain the same, and the pink color is consistent. The back of the cardigan features a clean, smooth design with no embellishments or logos. The fabric texture and sheen are identical to the front view.



Appendix Fig. 10. Generative AI image of a brown trench coat. (a) is the front image, and (b) is the back image.

A.10. A brown trench coat

- **Front Prompt:** A brown, water-repellent trench coat with a classic double-breasted design, featuring a subtle texture and a fitted silhouette. The coat has a high collar and a double vent at the back. The brown color is a rich, earthy tone with a slight sheen. Show the coat from the front, highlighting its clean lines and classic design.
- **Back Prompt:** The same brown, water-repellent trench coat from the front, now viewed from the back. The double vent and high collar are still prominent, and the fitted silhouette is evident. The texture and subtle sheen of the fabric are also visible from this angle. The brown color remains the same, with no distractions or obstacles in the background.

B. Examples of Incorrectly Generated Images

This section presents these main failure cases and discusses an in-depth analysis of the causes for each case, along with suggestions for future improvements.

B.1. A jeans (The model appears to be wearing the garments in the generated images)

- **Front Prompt:** Generate an image of a pair of high-waisted, dark blue, straight-leg jeans made from stretchy cotton blend fabric, featuring five pockets, including a small watch pocket, and a classic button fly closure. The jeans have a subtle fade effect with whiskering details and a slightly distressed hem.

hem.

- **Back Prompt:** Generate an image of the same pair of dark blue, straight-leg jeans from the back, showcasing the same stretchy cotton blend fabric, five pockets, and classic button fly closure. The jeans have a subtle fade effect with whiskering details and a slightly distressed hem, consistent with the front view.

Appendix Fig. 12 shows the generated front (a) and back (b) images for “high-waisted, dark blue, straight-leg jeans.” Both images were generated showing a virtual model wearing the jeans, rather than the garment itself. This outcome contradicts the intention of the system prompt, which instructed to “focus solely on the clothing, avoiding references to the model or background.”

Primary Causes for this phenomenon include, first, the training data bias of the diffusion models used (FLUX and SDXL). It is highly probable that common large-scale image datasets contain significantly more images of people wearing clothes than images of garments presented alone (e.g., flat lays, mannequin shots). Consequently, the model may have learned to strongly associate the concept of “clothing” with “clothing worn by a person.” This bias can predominantly influence the generation outcome, even when prompts explicitly instruct to exclude a model. Second, terms describing the fit or silhouette of the garment, such as “high-waisted” and “straight-leg,” might have acted as an ambiguity in prompt interpretation for the model, implicitly suggesting a human form. Furthermore, the strength of the negative instruction “exclude model” in the



Appendix Fig. 11. Generative AI image of a jeans. (a) is the front image and (b) is the back image.

system prompt might not have been sufficient to completely suppress this data bias or the strong association with certain keywords. Third, during the prompt generation process by the Llama3-Instruct model using CoT and Instruction Tuning, expressions naturally describing characteristics like “fit” or “silhouette” might have subtly signaled a human form to the image generation model. Lastly, if the front image was already generated with a model wearing it, the IP-Adapter, referencing this image, would likely maintain or reinforce this feature when generating the back image.

Future Improvement Suggestions include, firstly, considering methods to add or increase the weight of negative prompts with keywords that more clearly and forcefully instruct the absence of a model, such as “no human,” “garment only,” “flat lay photography,” or “product shot.” Additionally, further fine-tuning the diffusion models with a dataset composed of garment-only images (flat lays, mannequin shots, hanger shots, etc.) could guide the model to focus more on the intrinsic features of the clothing itself. Finally, a strategy is needed to refine the CoT and Instruction Tuning prompts so that the LLM minimizes the use of silhouette or fit-related terms that might imply a human form, and instead concentrates on the structural features, patterns, and texture of the garment.

B.2. A blue hoodie (The model appears to be wearing the garments in the generated images)

- Front Prompt: A blue hoodie with a soft, brushed cotton

fabric, featuring a bold, white graphic design on the chest, with a relaxed fit and a drawstring closure. The hoodie has a subtle sheen and a slight texture, with a ribbed cuff and hem.

- Back Prompt: A blue hoodie with a soft, brushed cotton fabric, featuring a bold, white graphic design on the back, with a relaxed fit and a drawstring closure. The hoodie has a subtle sheen and a slight texture, with a ribbed cuff and hem, showcasing the same design and fabric details as the front view.

Appendix Fig. 12 showing the generation results for a “blue hoodie with a soft, brushed cotton fabric,” similarly depicts a model wearing the garment in both the front (a) and back (b) views, akin to the jeans example in Appendix Fig. 11.

The Possible Causes for this result are largely the same as those discussed for Appendix Fig. 11. That is, it is highly likely that training data bias led the model to strongly associate the concept of a “hoodie” with a person wearing it. Furthermore, silhouette-related expressions in the prompt, such as “relaxed fit,” might have implicitly suggested a human form, and the constraints of the negative prompt might not have been sufficiently strong. The subtle phrasing differences in the LLM-generated prompt or the IP-Adapter reflecting a model’s presence from an already generated front image cannot be ruled out.

Therefore, Future Improvement Directions can also be approached similarly to those for Appendix Fig. 11. Increasing the clarity and strength of negative prompts and fine-tuning the diffusion model with data of model-free garment product imag-



Appendix Fig. 12. Generative AI image of a blue hoodie. (a) is the front image and (b) is the back image.



Appendix Fig. 13. Generative AI image of a mint windbreaker. Both (a) and (b) show the front.

es could be effective. Additionally, it is important to modify the prompt generation strategy to minimize human-related implications in the prompts generated by the LLM.

B.3. A mint windbreaker (All of them depict only the front view)

- Front Prompt: A mint green windbreaker with a smooth, water-repellent nylon fabric, featuring a sleek and modern design with a zip-up closure, adjustable cuffs, and a fitted silhouette. The jacket has a subtle sheen and a soft, velvety texture.

The color is a bright, calming mint green with a slight blue undertone.

- Back Prompt: The back of the mint green windbreaker showcases the same smooth, water-repellent nylon fabric, with the same sleek and modern design elements, including the zip-up closure and adjustable cuffs. The color remains a bright, calming mint green with a slight blue undertone, and the fabric has the same subtle sheen and soft, velvety texture as the front.

Appendix Fig. 13 displays the generation result for a “mint

green windbreaker.” Despite requesting a back view image (b), an image nearly identical to the front view image (a) was generated, or the back-specific features were not properly represented, effectively regenerating a front view.

The Primary Cause of this issue can be attributed to LLM’s prompt generation error. Specifically, the back-view prompt for Appendix Fig. 13 includes the phrase “including the zip-up closure and adjustable cuffs.” A “zip-up closure” is typically a prominent feature on a garment’s front. The inclusion of such an explicit front-facing feature in the prompt describing the back view likely confused the image generation model (SDXL + IP Adapter), leading it to regenerate a front view. Second, the incompleteness of the CoT logic, intended to maintain consistency between front and back views, might also be a factor. Instead of clearly instructing the unique features of the back, the logic might have simply repeated front features, as suggested by phrases like “the same ... design elements,” or lacked the logic to explicitly exclude front elements that should not appear on the back. Third, the influence of the IP-Adapter, which references the front image, might have been excessive, causing it to prioritize replicating the visual features of the front image over the textual instructions of the back-view prompt. Lastly, the diffusion model itself might have a bias towards generating front views for certain garment items (e.g., windbreakers) if it was trained on more front-view images, especially when given ambiguous or conflicting prompts.

To address these issues, future improvement suggestions include, first, thoroughly checking and revising the LLM prompt generation logic. The LLM’s CoT and Instruction Tuning logic must be improved to explicitly exclude features present on the front but not on the back (e.g., front zippers, specific front pocket placements) when generating back-view prompts. Simultaneously, prompts should be constructed to more specifically and clearly emphasize unique back features (e.g., back seams, hood attachment details, center back prints, yoke designs), and phrasing that clarifies the viewpoint and exclusions, such as “the back view should clearly show...”, “from the rear perspective...”, “excluding front-specific details like the main zipper,” should be reinforced. Second, one could consider adjusting the IP-Adapter’s weights by experimentally balancing the influence of the image prompt from the IP-Adapter and the text prompt on the back-view generation, ensuring that textual instructions are adequately reflected. Third, adding a prompt consistency, and clarity verification step could be considered. This step would automatically check if the LLM-generated front/back prompt pair contains contradictory elements in the

back prompt or if the back view is clearly instructed, using rule-based or model-based validation.

C. Source Code

```
reasoning_prompt = [
    {"role": "system", "content":
        # Role Prompt
        "You are an expert prompt engineer specializing in generating detailed and effective prompts for image generation models."
        "The focus should be on the clothing details, such as fabric, design, and craftsmanship."
        "Avoid including the model, background, or pose in the descriptions."
        "Your task is to create prompts that request images from multiple angles: front, and back."
        "Ensure the descriptions for both the front and back are coherent, consistent, and emphasize the same clothing attributes."},

    {"role": "user",
     "content":
        f"""### instruction ###
        I want to generate images of garment from two different angles: front, and back. Ensure that the garment is the same in both angles, and the design, colors, and details of the clothing remain consistent.
        Please ensure there are no obstacles in the background, allowing only the clothing to be visible.
        Focus solely on the clothing, avoiding references to the model or background. Ensure that the prompts for each angle are clear and concise, and do not exceed 77 tokens.

        ### Thought Process ###
        1. Identify the main features of the garment: color, fabric type, design details.
        2. Create a prompt for the front angle, emphasizing key attributes.
        3. Create a prompt for the back angle, ensuring consistency with the front description.

        ### Input Text ### Query: {user_query}

        =====
        New Prompt:
        """]
```